

09 — Second-order circuits: series RLC

MTE syllabus item 7 (under-, over-, critically damped, undamped) · not yet in the class notes: the question is a textbook problem · needs 08 ($0^-/0^+/\infty$, Laplace)

Map

1 Why RLC is second order

→ 2 The characteristic equation

→ 3 α and ω_0 decide everything

→

4 The four cases

→

5 Using the initial conditions

The question this file answers

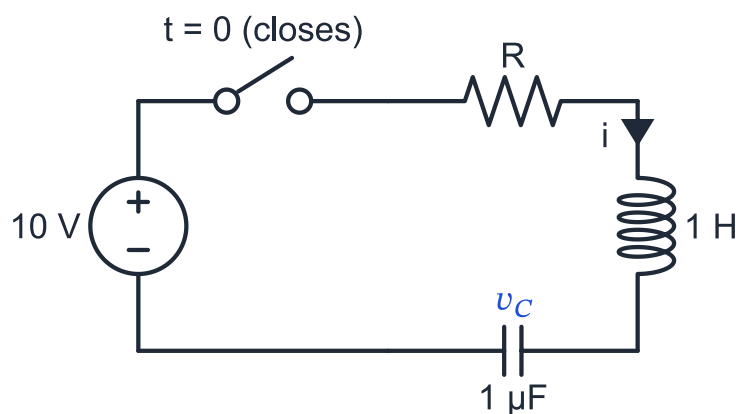
- **textbook** 10 V is switched onto a series R, 1 H, 1 μF circuit at $t = 0$ (everything at rest). Find $v_C(t)$ for $R = 2.5 \text{ k}\Omega$, 2 $\text{k}\Omega$, 1.2 $\text{k}\Omega$ and 0. Classify each.

The class notes (set 2 p.18) already mark the boundary: the one first-order formula is "not for RLC and LC circuits". This file is what replaces it. Facts from the circuits knowledge base (Hayt; Van Valkenburg; Alexander & Sadiku ch. 8); every curve below was checked by simulating the circuit.

Build

TEXTBOOK

Find $v_C(t)$ after the switch closes, for $R = 2.5 \text{ k}\Omega$, 2 $\text{k}\Omega$, 1.2 $\text{k}\Omega$ and 0.



$L = 1 \text{ H}$, $C = 1 \mu\text{F}$, source 10 V, switch closes at $t = 0$. $i(0^-) = 0$, $v_C(0^-) = 0$.

GUESS FIRST

Make R very large, then zero. What does v_C do in each case? Can it ever go **above** 10 V?

1 · Why it is second order

The circuit has **two** energy stores: the inductor (current) and the capacitor (voltage). Energy can pass back and forth between them, like a mass on a spring. The resistor is the friction.

KVL round the loop, with $i = C \frac{dv_C}{dt}$:

$$L C \frac{d^2 v_C}{dt^2} + R C \frac{dv_C}{dt} + v_C = V$$

A second derivative: one time constant is no longer enough.

2 • The characteristic equation

By Laplace (08 §6), with everything at rest: $I(s) = \frac{V/s}{R + sL + 1/sC}$. Multiply top and bottom by s/L :

$$I(s) = \frac{V/L}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

The denominator set to zero is the **characteristic equation**. Its roots (the poles) decide the shape of the response:

$$s^2 + 2\alpha s + \omega_0^2 = 0 \quad s = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

3 • Two numbers decide everything

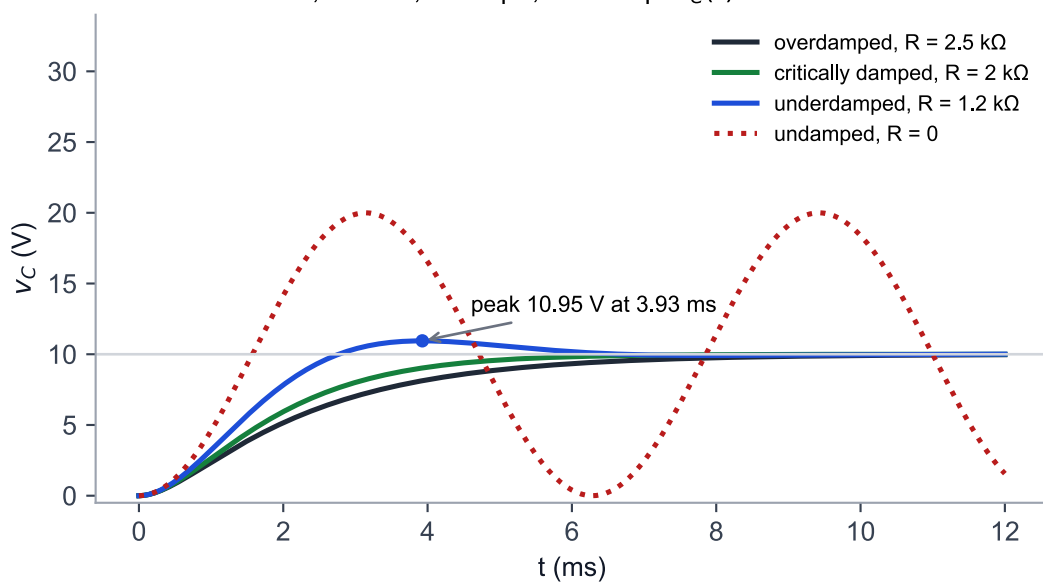
Symbol	Name	Series RLC	Here ($L = 1 \text{ H}$, $C = 1 \text{ } \mu\text{F}$)
α	damping coefficient (neper frequency)	$\frac{R}{2L}$	$\frac{R}{2}$ per second
ω_0	undamped natural frequency	$\frac{1}{\sqrt{LC}}$	$\frac{1}{\sqrt{(10^{-6})}} = \mathbf{1000 \text{ rad/s}}$
ζ	damping ratio	$\frac{\alpha}{\omega_0}$	

Compare α with ω_0 . The **critical resistance** is where $\alpha = \omega_0$: $R_c = 2\sqrt{\frac{L}{C}} = 2\sqrt{(10^6)} = \mathbf{2 \text{ k}\Omega}$.

4 · The four cases

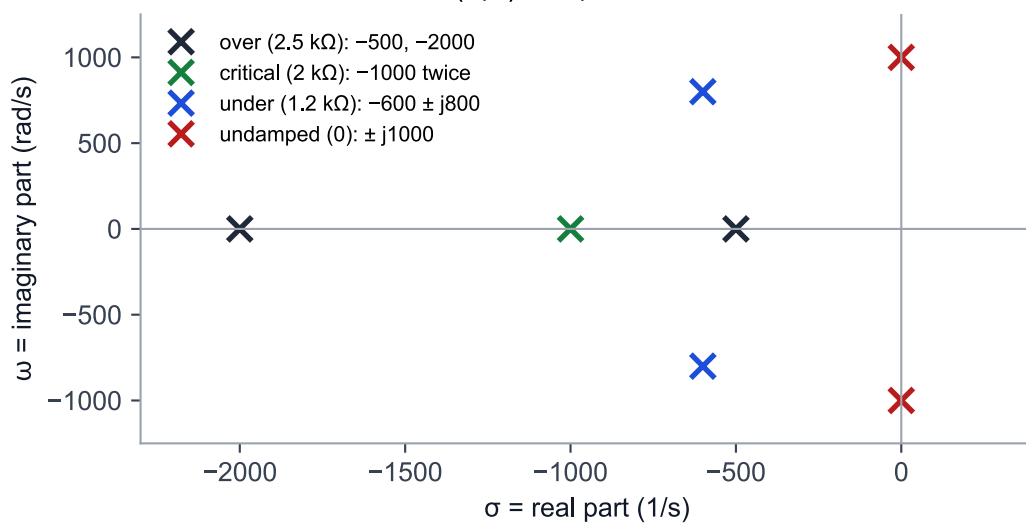
R	α	Roots	Case	$v_C(t)$ (V)
2.5 k Ω	$1250 > \omega_0$	-500, -2000 (real, different)	overdamped	$10 - \frac{40}{3}e^{-500t} + \frac{10}{3}e^{-2000t}$
2 k Ω	$1000 = \omega_0$	-1000, twice (real, equal)	critically damped	$10 - 10(1 + 1000t)e^{-1000t}$
1.2 k Ω	$600 < \omega_0$	$-600 \pm j800$ (complex)	underdamped	$10 - e^{-600t}(10 \cos 800t + 7.5 \sin 800t)$
0	0	$\pm j1000$ (imaginary)	undamped	$10(1 - \cos 1000t)$

Series RLC, L = 1 H, C = 1 μ F, 10 V step: $v_C(t)$ for four values of R



Each formula was checked against a time simulation of the circuit at $t = 1, 2, 3.93$ and 5 ms; they agree to 6 digits.

Roots of $s^2 + (R/L)s + 1/LC = 0$ for the four cases



Where the roots sit. On the real axis: no oscillation. Off it: oscillation at the imaginary part. The further left, the faster the decay.

What each case looks like:

- **Overdamped** ($\alpha > \omega_0$, $\zeta > 1$): two decaying exponentials; **slow**, no overshoot. The slow root (-500 , i.e. $\tau = 2$ ms) dominates.
- **Critically damped** ($\alpha = \omega_0$, $\zeta = 1$): the form $(A + Bt)e^{-\alpha t}$; the **fastest rise with no overshoot**.
- **Underdamped** ($\alpha < \omega_0$, $\zeta < 1$): a decaying oscillation $e^{-\alpha t} \cos(\omega_d t + \phi)$, at the **damped frequency** $\omega_d = \sqrt{(\omega_0^2 - \alpha^2)} = \sqrt{(1000^2 - 600^2)} = 800$ rad/s. It **overshoots**: peak **10.95 V** at $t = \pi/\omega_d = 3.93$ ms.
- **Undamped** ($R = 0$, $\alpha = 0$): the energy swings between L and C forever at ω_0 ; v_C goes up to **20 V**, twice the source.

The Guess answer: large $R \rightarrow$ slow creep up to 10 V; $R = 0 \rightarrow$ endless oscillation between 0 and 20 V. So yes, v_C **can** exceed the source voltage, whenever the circuit is underdamped.

CHECK 1

Series RLC with $L = 10$ mH, $C = 1$ μ F. (a) Find ω_0 and the critical R . (b) Classify $R = 100 \Omega$ and $R = 500 \Omega$.

5 • Using the initial conditions

The table's constants come from the two initial conditions (08: what can't jump):

- $v_C(0^+) = v_C(0^-) = 0$ (capacitor voltage cannot jump).
- $i(0^+) = i(0^-) = 0$ (inductor current cannot jump), and since $i = C \frac{dv_C}{dt}$, this gives $\frac{dv_C}{dt}(0^+) = 0$.
- Final value: at ∞ the capacitor is open, the inductor a short, so $v_C(\infty) = 10$ V.

Overdamped, worked: $v_C = 10 + A_1 e^{-500t} + A_2 e^{-2000t}$.

$$v_C(0) = 0: A_1 + A_2 = -10 \qquad \frac{dv_C}{dt}(0) = 0: -500A_1 - 2000A_2 = 0$$

The second gives $A_1 = -4A_2$, so $-3A_2 = -10$: $A_2 = \frac{10}{3}$, $A_1 = -\frac{40}{3}$, as in the table.

Underdamped current, the other quantity asked most often: $i(t) = \frac{V}{\omega_d L} e^{-\alpha t} \sin \omega_d t = 12.5 e^{-600t} \sin 800t$ mA (simulation at 1 ms: 4.92 mA $\checkmark$).

CHECK 2

Why must the current in the undamped case be $i(t) = 10 \sin(1000t)$ mA? Check it with $i = C \cdot dv_C/dt$.

Exam form

Series RLC: characteristic equation $s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$; $\alpha = \frac{R}{2L}$, $\omega_0 = \frac{1}{\sqrt{LC}}$, $\zeta = \alpha/\omega_0$, roots $s = -\alpha \pm \sqrt{(\alpha^2 - \omega_0^2)}$.

Condition	Case	Form of the natural response
$\alpha > \omega_0$ ($R > 2\sqrt{L/C}$)	overdamped	$A_1 e^{s_1 t} + A_2 e^{s_2 t}$
$\alpha = \omega_0$ ($R = 2\sqrt{L/C}$)	critically damped	$(A_1 + A_2 t) e^{-\alpha t}$
$\alpha < \omega_0$	underdamped	$e^{-\alpha t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t)$, $\omega_d = \sqrt{(\omega_0^2 - \alpha^2)}$
$\alpha = 0$ ($R = 0$)	undamped	$A_1 \cos \omega_0 t + A_2 \sin \omega_0 t$

Step response = final value + natural response. Constants from $v_C(0^+)$ and $i_L(0^+)$ (i.e. $dv_C/dt(0^+) = i_L(0^+)/C$).

Parallel RLC (for recognition): same ω_0 , but $\alpha = \frac{1}{2RC}$, so a **small** R gives heavy damping, the opposite of series.

Traps

- Using $x(t) = x(\infty) + [x(0^+) - x(\infty)]e^{-t/\tau}$ on an RLC circuit (set 2 p.18 warns against it).
- Writing $\alpha = R/L$ instead of $R/2L$.
- Oscillating at ω_0 in the underdamped case: the oscillation is at ω_d , which is smaller.
- Forgetting the second initial condition ($dv_C/dt(0^+)$ from the inductor current).
- Parallel RLC: using the series α .

Self-test

1. Write the characteristic equation of a series RLC and define α and ω_0 .
2. $L = 0.1$ H, $C = 10$ μ F. Find the critical resistance.
3. For $R = 1.2$ k Ω in the question, what is the period of the ringing?
4. Which case settles fastest without overshoot?
5. Why can an LC circuit ($R = 0$) never reach a steady state?

Answers

Check 1. (a) $\omega_0 = 1/\sqrt{(10^{-2} \times 10^{-6})} = \mathbf{10\,000\ rad/s}$; $R_C = 2\sqrt{(10^{-2}/10^{-6})} = \mathbf{200\ \Omega}$. (b) $100\ \Omega$: underdamped ($\alpha = 5000 < 10\,000$). $500\ \Omega$: overdamped ($\alpha = 25\,000 > 10\,000$).

Check 2. $i = C\,dv_C/dt = 10^{-6} \times 10 \times 1000 \sin 1000t = 0.01 \sin 1000t\ \text{A} = \mathbf{10\ sin(1000t)\ mA}$. It starts at 0, as the inductor requires.

Self-test.

1. $s^2 + (R/L)s + 1/LC = 0$; $\alpha = R/2L$, $\omega_0 = 1/\sqrt{LC}$.

2. $R_C = 2\sqrt{(0.1/10^{-5})} = 2 \times 100 = \mathbf{200\ \Omega}$.

3. $T = 2\pi/\omega_d = 2\pi/800 = \mathbf{7.85\ ms}$.

4. **Critically damped.**

5. No resistor to turn the energy into heat: it keeps passing between L and C, so the roots sit on the $j\omega$ axis and never decay.